

Cross-Architecture Phenomenon Mapping (Week 4)

Note: Token collapse is about the data points (tokens) merging together, while dimensional reduction is about the underlying space (features) shrinking.

Note: I have made a distinction between Computational Complexity and Circuit Complexity. Previous sources have used "computational complexity" to address the minimum size and depth of physical or abstract logic gates required to compute a specific function. In this segment, this idea will be termed as circuit complexity. On the other hand, computational complexity will refer to time complexity.

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Phenomenon	Transformer	Mamba / SSM	Diffusion LLM	Mixture-of-Experts (MoE)
Phenomenon	Transformer	Mamba / SSM	Diffusion LLM	Mixture-of-Experts (MoE)
Token clustering / collapse	Well-studied: Token collapse is an inherent, energy-minimizing pathology of self-attention. Mathematically, attention acts as a Wasserstein gradient flow on a hypersphere, naturally dragging all tokens into a uniform Dirac mass (Rigollet, 2025). Before total collapse, however, tokens form expressive semantic clusters (Rigollet, 2025). Transformers preserve these clusters by utilizing layer normalization as a "speed regulator" (Karagodin et al., 2025). While Post-LN maintains a constant velocity that accelerates collapse, Pre-LN and Peri-LN configurations allow token magnitudes to grow (Karagodin et al., 2025). This growth dynamically decays particle velocity, effectively braking the system and freezing tokens inside their useful multi-clustered states to maintain representation rank (Karagodin et al., 2025)	Unknown: It remains unknown if Mamba's sequential processing creates an analogous mathematical collapse.	Unknown: It is currently unknown whether tokens cluster or collapse into sub-manifolds during denoising steps. While there is existing literature on diffusion models, it cannot be safely generalised for diffusion LLMs.	Partially studied (Expert Collapse): <i>Grassmannian Mixture-of-Experts</i> proves that standard Euclidean routing causes tokens to flood identical experts, but mapping to a Grassmannian manifold mathematically prevents this by forcing experts into orthogonal subspaces.

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Phenomenon	Transformer	Mamba / SSM	Diffusion LLM	Mixture-of-Experts (MoE)
Dimensional reduction	Well-studied: <i>Bridging the Dimensional Chasm</i> shows a geometric framework where tokens compress from high-dimensional embedding spaces (10^3) onto ≈ 10 -dimensional submanifolds representing human semantics (Song et al. 2025). Such phenomenon is evident in effective LLM models.	Partially Studied: Mamba compresses past tokens into a fixed-size recurrent hidden state matrix. When sequence lengths vastly exceed the model’s training threshold, the model encounters a memory capacity ceiling (Chen et al., 2024).	Partially Studied: In deep Diffusion Transformers, researchers have identified a structural vulnerability called a mean-dominated collapse state, triggered by a phenomenon known as Mean Mode Screaming (MMS). When scaling DiTs (Diffusion Transformer) to extreme depths (400 to 1000 layers) for video or high-fidelity image generation, the residual stream becomes progressively dominated by the pure-mean component. The centered variation component gets mathematically squeezed out ($\ c(X)\ _F \rightarrow 0$). The tokens experience a severe dimensional reduction, homogenizing into a uniform, low-variance state that completely suppresses visual variation and crashes generation quality.	Unknown: It remains an open gap how dimensional reduction impacts routing efficiency or if routed tokens change their intrinsic dimension.

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Phenomenon	Transformer	Mamba / SSM	Diffusion LLM	Mixture-of-Experts (MoE)
Information compression	3-Phase Pattern: Llano et al.(2025) outlines a "Mix-Compress-Refine" pattern where intermediate layers squish information into a dense, high-energy summary. Such observations attempts to explain how transformer LLMs organize their computation in depth by controlling attention and representational compression via massive activations	Multi-Scale Filtering: <i>Understanding Input Selectivity in Mamba</i> models Mamba as actively counteracting memory decay by deliberately holding onto critical tokens and aggressively forgetting useless ones (Huang et al., 2025).	Iterative Semantic Locking: An example of diffusion LLM, <i>Simple and Effective Masked Diffusion Language Models</i> , proves continuous semantic certainty increases through reverse prediction, yielding a tighter ELBO (Sahoo, 2024).	Complexity Gradients: <i>DynaMoE</i> links token information entropy to compute, dropping simple tokens (0 experts) while heavily routing complex tokens (4-5 experts) (Gulmez, 2026).

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Phenomenon	Transformer	Mamba / SSM	Diffusion LLM	Mixture-of-Experts (MoE)
Representational rank	Rank decreases with depth: Without non-linearities (like MLPs) or skip-connections, token representations rapidly homogenize across successive layers. The token matrix $X \in \mathbb{R}^{n \times d}$ experiences complete rank collapse, where all singular values except the first one drop to zero ($\sigma_2(X) \rightarrow 0$). The tokens collapse into a uniform, $rank - 1$ subspace (Dong et al., 2021).	Similarity to transformer: According to Yang et al. (2025), Mamba’s token-mixing framework acts as a structural low-rank approximation of traditional Softmax Attention. Unlike Transformers, which possess the geometric freedom to construct full-rank token matrices, Mamba’s linear recurrence inherently restricts its feature space to a lower matrix rank. This structural bottleneck means Mamba has a weaker absolute representational capacity than standard self-attention. However, it scales linearly and retains a significantly higher matrix rank and expressivity than pure Linear Attention, positioning Mamba as a theoretically optimal, high-efficiency middle ground for sequence modeling.	Unknown: Literature does not explicitly cover representational rank limits in diffusion structures.	Unknown: It is unknown how rank evolves with depth in routed models.

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Phenomenon	Transformer	Mamba / SSM	Diffusion LLM	Mixture-of-Experts (MoE)
Circuit complexity	TC⁰ Class: Can compute polynomial precision functions via $\mathcal{O}(n^2)$ interactions.	TC⁰ Class: <i>The Computational Limits of State-Space Models</i> proves Mamba is in the exact same TC^0 complexity class as Transformers, matching their expressivity despite sequential processing (Chen, 2024).	Unknown: Formal computational bounds for discrete language diffusion generation are not covered.	Varies: Complexity depends inherently on the gating function and expert assignment structure.
Computational Complexity	$O(T^2d^2)$ training , but dominates to $O(T^2)$ as sequence length grows. Inference: $O(Td + d^2)$	$O(TdN)$ training , constant $O(1)$ step-time complexity ($O(d \cdot N)$ math ops) and a flat, constant memory footprint that does not increase regardless of whether the model processes 100 or 1,000,000 tokens (Chadha, 2026).	No standard training time complexity. For a Transformer-backed diffusion model, the computational complexity scales as $O(K \cdot T^2d)$ across K sampling loops (Wu et al., 2026).	$O(T^2d)$ training inference time complexity of a Mixture of Experts (MoE) layer is practically identical to a dense layer of equivalent active parameters: $\mathcal{O}(k \cdot d_{model} \cdot d_{ffn})$, where k is the number of experts activated per token (typically 2), d_{model} is the hidden dimension, and d_{ffn} is the expert’s inner layer dimension (Takashiro, 2026)

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Phenomenon	Transformer	Mamba / SSM	Diffusion LLM	Mixture-of-Experts (MoE)
Implicit vs. Explicit Interaction	Explicit Attention: Uses explicit all-to-all dense interactions per layer ($\mathcal{O}(n^2)$).	Implicit Micro-Routers: <i>The Hidden Attention of Mamba Models</i> shows Mamba dynamically generates implicit attention matrices on the fly, using 3 orders of magnitude more "hidden" micro-routers than Transformer heads (Ali, 2024).	Bidirectional Context: <i>Large Language Diffusion Models</i> shows all tokens update simultaneously during inference, continuously refining confidence globally without left-to-right masks.	MoEs natively execute explicit attention at the sequence layer level to capture token-to-token contextual relationships (Zhang et al., 2025). However, the sparse routing mechanism introduces a macro-level implicit parameter-token interaction graph

Variable Key: T = Sequence Length, d = Model Hidden Dimension, N = SSM State Dimension, K = Denoising Steps, E = Total Experts, k = Active Experts per token.

Gaps Identified for Section 12 (Open Problems)

1. **Collapse Theory Beyond Transformers:** While Rigollet’s theorem and ODE models explicitly prove representations inevitably collapse in deep Transformers, no such proof or disproof exists for Mamba or Diffusion architectures.
2. **Manifold Traversal and Dimensional Reduction:** Dimensionality strictly compresses to ≈ 10 -dim semantic submanifolds in Transformers, but it is entirely unknown if the SSM hidden state or the diffusion denoising trajectory exhibits this same geometric expansion-contraction.
3. **Representational Rank Evolution:** The phenomenon of decreasing representational rank due to massive activations (Attention Sinks) is isolated to pre-norm Transformers. How rank evolves in sequential SSMs or sparse MoEs remains a critical gap.
4. **Routing Geometry vs. Intrinsic Dimension:** While *Grassmannian Mixture-of-Experts* maps expert routing to curved manifolds, there is no existing literature connecting these routing decisions directly to a token’s changing intrinsic dimension.

Literature Synthesis: Agreements and Contradictions

- **Agreement on Transformer Collapse:** Multiple distinct analytical lenses converge on the same vulnerability. *The Mean-Field Dynamics of Transformers* uses geometric/manifold analysis to prove collapse on $\mathbb{S}^{d-1}$, while *Flowing Through Layers* reaches the exact same conclusion using continuous dynamical systems (ODEs) proving contractive pull.
- **Agreement on Cross-Architecture Expressivity:** *The Computational Limits of State-Space Models* formally bridges Transformers and Mamba by proving they occupy the exact same complexity class (TC^0). Furthermore, *The Hidden Attention of Mamba Models* supports this equivalence mechanically, proving Mamba executes a highly factored, implicit form of self-attention.
- **Nuance/Contradiction on the Cause of Representation Collapse:** *Attention Sinks and Compression Valleys* explicitly attributes token representation collapse to massive token activations in the residual stream interacting with LayerNorm. In contrast, *The Mean-Field Dynamics of Transformers* argues that collapse is an intrinsic mathematical destiny caused strictly by the softmax attention matrix averaging mechanism, existing independently of specific architectural quirks like residual streams or massive activations.